

Module 14) Python – Collections, functions and Modules

5. Accessing Tuples

- **Accessing tuple elements using positive and negative indexing.**

1. Introduction to Tuple Indexing

Tuples are ordered sequences, meaning each element has a fixed position. Python provides two indexing methods:

- Positive indexing (left to right, starting at 0)
- Negative indexing (right to left, starting at -1)

Both methods allow $O(1)$ time complexity for element access.

2. Positive Indexing

Characteristics

- Starts at 0 for the first element
- Increases by 1 for each subsequent element
- Last element at index $\text{len}(\text{tuple}) - 1$

3. Negative Indexing

Characteristics

- Starts at -1 for the last element
- Decreases by 1 moving leftward

- First element at index `-len(tuple)`

- **Slicing a tuple to access ranges of elements.**

1. Introduction to Tuple Slicing

Tuple slicing extracts a subsequence (sub-tuple) from a tuple using the syntax:

`tuple[start:stop:step]`

- Returns a new tuple (original remains unchanged)
- Works similarly to list slicing but produces immutable results

Indexing Systems

Slicing works with both indexing schemes:

- **Positive indices:** Count from left (0 = first element)
- **Negative indices:** Count from right (-1 = last element)

Special Forms

Slice Pattern	Result	Use Case
<code>tuple[:]</code>	Full shallow copy	Duplicate immutable data
<code>tuple[::2]</code>	Every other element	Downsampling
<code>tuple[::-1]</code>	Reversed tuple	Reverse iteration
<code>tuple[-3:]</code>	Last 3 elements	Tail operations

Performance Characteristics

- **Time Complexity:** $O(k)$ where k is slice size
- **Space Complexity:** $O(k)$ for the new tuple
- **Memory:** Shares references to original elements (shallow copy)